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Soft Actor-Critic Reinforcement Learning Approach to Multi-Drone 3D Terrain Scanning and Target Detection in Search and Rescue Operations

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I. Nomenclature

a	=	action vector chosen by a drone at the current time step (Δx , Δy , Δz)
a'	=	action that will be sampled by the policy in the next state s'
α	=	Temperature coefficient that balances exploration vs. exploitation in SAC
D	=	replay buffer that stores experience tuples ⟨*s, a, r, s'*⟩ for off-policy learning
γ	=	discount factor that weights future rewards when computing target Q-values
r	=	immediate reward obtained after executing action a in state s
s	=	current state
s'	=	next state
t	=	time index during navigation
y	=	Target Q-value used to train the critic network, computed from r, s′, a′ and γ
X,Y,Z	=	Drone Cartesian coordinates in the 100 × 100 × 10 3-D search space (with Z = altitude)
UAV	=	Unmanned Aerial Vehicle
SAC	=	Soft Actor-Critic
RL	=	Reinforcement Learning
ML	=	*Machine Learning*
SAR	=	Search and Rescue

II. Abstract

Unmanned Aerial Vehicle use has increased in many civil operations, such as search and rescue missions. As these vehicles become essential in more applications, the need to control and coordinate swarms of them efficiently increases. Search and rescue missions are usually carried out in difficult, dynamic, and complex scenarios where the chances of success rely on how well and fast the unmanned vehicles navigate and interact with these environments. Reinforcement Learning has been widely used in training UAVs for different tasks, as they require less prior knowledge than other Machine Learning methods of their environment. This paper's main contribution is the use of the recent Soft Actor-Critic method to train swarms of drones in terrain scanning and target detection through increasingly complex terrains, from sparse scenarios to densely populated obstacle fields. Unlike other methods, SAC encourages exploration and adaptability through an entropy term, making it suitable for target location in unknown areas. The model is trained in a simulated environment with a reward system that prioritizes drone coordination to cover new areas and find the most targets, while avoiding obstacles and area limits. Sensing capabilities of drones vary according to their flight altitude, so they learn to balance between larger area coverage and more precise detection. The model is then compared to other navigation algorithms, and it shows that it can be more efficient than them on average, finding 58% of targets in the first hour, and 75% of them in the second hour. Out of the other path strategies explored, Random Walk showed the best results, finding targets with 46% and 73% success in the first and second hours, respectively. The preliminary results show that the model can be further improved, since the model does not enhance operations significantly at its current state. Future research can focus on implementing Soft-Actor Critic with

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other algorithms for hybrid approaches that enhance performance; training and testing the model in new environments, including fault detection systems in drones, or testing other operations that are not the target location.

Keywords: *Autonomous Navigation, Path Planning for Multiple Mobile Robots, Aerial Robotics, Collision Avoidance, Soft-Actor Critic, Search and Rescue, Adaptive Systems*

III. Introduction

In recent years, Unmanned Aerial Vehicles (UAVs) have emerged as a transformative technology for many civil applications due to their low maintenance cost, high mobility, ease of deployment, and ability to explore areas by hovering [1]. These applications include real-time monitoring, goods delivery, precision agriculture, infrastructure inspection, and SAR operations [2]. In SAR operations, UAVs help inspect limited-access, harsh, and dynamic areas, reducing the effort and time required to analyze affected zones. Consequently, human responders can focus on other managerial strategic tasks [3]. Additionally, UAVs mitigate the risk of having a human onboard operator in these complex environments and situations [4].

UAVs are usually deployed in swarms, as they can outperform individual drones by covering larger areas in less time, supporting different roles and objectives, and increasing operational adaptability and robustness [5]. Despite the advantages, coordinating a variety of drones in a 3-D obstacle-rich environment is challenging. Each drone must respond to the dynamic environment and its topography, avoiding collisions with both static and dynamic obstacles, including other drones, while contributing to the swarm's collective objectives. Since one of the primary objectives is to cover the selected area in the shortest time possible, there is substantial research on path strategies, drone collaboration, and drone technology [6].

Reinforcement Learning (RL), a form of Machine Learning (ML), trains agents to solve a problem by interacting with their environments and maximizing the cumulative rewards. Unlike other ML methods, RL does not require prior knowledge of the environment [7]. Soft Actor-Critic (SAC) is a Deep Reinforcement Learning (DRL) learning method that does not require any modeling of the environment, uses neural networks to control high-dimensional state and action spaces, and encourages agent exploration through its entropy term. In the realm of UAV mission planning, these stochastic strategies offer greater adaptability than deterministic strategies [8].

This paper proposes a SAC-trained approach for swarms of UAV drones conducting SAR operations. The model will be trained and tested across 3D environments of increasing complexity, with the objective of detecting missing targets. The drones will have different sensing accuracies and ranges depending on flight altitude.

IV. Background

A. Machine Learning in Search and Rescue

ML has experienced significant growth in popularity due to its versatility and performance across many applications. SAR missions have benefited from ML methods by being able to automate many processes, such as target location prediction, area reconnaissance, optimizing resource allocation, and supply delivery. For target prediction, the authors in [9] explore supervised and unsupervised ML approaches to overcome GPS limitations in indoor environments. To avoid systems tracking agents similar to the targets, the authors in [10] propose a new NN method that labels and tracks all agents in sight, with labels propagating frame-to-frame. In [11], an extensive survey of Deep Learning methods for predicting target location based on the acoustics they emit is presented. To better sense and understand the environment, many ML models are being developed for image recognition. For example, Auto-encoders have been used for image classification of forest areas [12], Generative Adversarial Networks have been trained to enhance object detection in low-resolution images [13], and Visual Transformers and Convolutional Neural Networks (CNN) have been developed with diverse datasets to identify objects in different domains and scenarios [14, 15].

Biologically inspired swarm formations have been used for multi-UAV operations that allow them to assemble and move using decentralized algorithms [16]. Other efforts have explored different swarm formations based on distance constraints [17]. For area reconnaissance UAVs, it is stated in [18] that ML-backed systems increase the chances of mission success drastically. Recent studies have explored new swarm architectures with the use of Genetic Algorithms and Neural Networks (NN) [19] or the combination of biologically inspired designs with NN, as seen in the pheromone diffusion model presented in [20].

For resource allocation optimization after COVID-19, researchers in [21] have developed ML models to identify the areas in most economic distress based on local phone subscriptions and send strategic aid to combat poverty. CNN and Long Short-Term Memory (LSTM) models have also been used for resource and energy management [22].

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B. Reinforcement Learning in Search and Rescue

Due to their ability to work with less prior knowledge of the environments than other methods, RL methods have been studied for different SAR scenarios. In maritime SAR, RL methods have proven useful for path planning, achieving positive results without a predetermined navigation plan [23] for developing cooperative UAV and Unmanned Surface Vehicle (USV) systems using DRL and CNNs [24]; for improving the chances of finding people in the water after a man overboard incident [25], or a hybrid approach using surface buoys, sensor nodes, and USVs for underwater target location [26].

In forest areas, RL has been used to map and plan SAR missions under the forest canopy, leveraging a cooperative approach between a sensing swarm of UAVs and a central ground station to overcome challenges such as unreliable GPS and drone-to-drone communication [27], or to predict and track fire propagation [28]. RL has also been combined with nature-inspired algorithms to improve operations, as seen in [29], where Q-Learning is combined with the Flower Pollination algorithm to accelerate solution convergence and adaptability to both dynamic and static 2-D grids.

In urban environments, a DRL with both CNN and LSTM layers has proven helpful for robot navigation [30]. Hierarchical RL has been studied to allow robot teams to learn from their surroundings, past behaviors, experiences from other team members, and the expert knowledge of human operators [31]. Other research has developed innovative DRL methods to improve SAR operations during urban floods [32].

C. Machine Learning in Search and Rescue

SAC RL is relatively new to SAR, with UAVs, with the method being first introduced in 2018 [33]. One study uses SAC to enhance energy use and consumption in UAV missions on an established tracking system [34]. In [35], SAC is used to train humanoid NAO robots to plan their trajectories and interact with the environment objects. Additionally, SAC has been shown to outperform other algorithms in optimizing task offloading and resource allocation [36]. Practical experiments with laser sensors have used SAC in combination with various methods for obstacle avoidance and navigation [37]. Compared with Proximal Policy Optimization and Twin Delayed Deep Deterministic, SAC has faster convergence, better path smoothing, and overall better performance [38].

Research is gaining traction, with most papers published in recent years. Still, to the best of the authors' knowledge, SAC has not been implemented for 3D obstacle avoidance and path planning in unknown environments for UAV swarms in SAR missions with multiple targets.

V. Methodology

The approach to demonstrate the usefulness of SAC methods for swarm UAV SAR operations is to create multiple environments of varying complexity and to design the SAC RL method details, including the model's state representation, action space, and reward function components.

A. Simulation Environment

In this project, the environment is a 3-dimensional 100 x 100 x 10 space, representing the area of interest in the SAR mission. In this space, targets will be placed randomly on the ground, and drones will operate in the $z \in [1,10]$ range. To test scenarios of different difficulty and complexity, ten maps are created of increasing object density from 0% to 45% of the area covered by obstacles, as seen in Fig. 1. These obstacles are placed randomly on the canvas and are 3-dimensional as well, with a maximum height of 8 units and a maximum width of 5 units. Obstacles block drones' movement and sight, meaning drones need to avoid them by going around or above in order to search the map, just like they would in a real scenario such as an urban area.

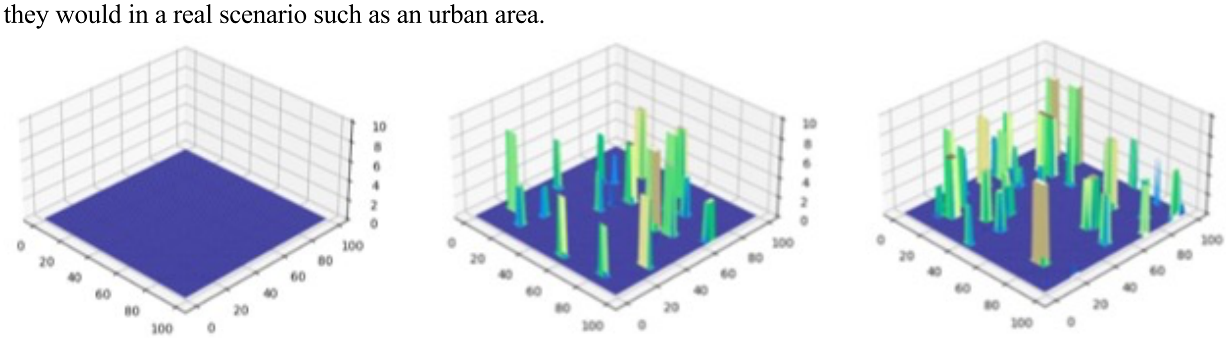

Figure 1: Three maps of increasing complexity, with 0%, 25%, and 45% obstacles in the environment.

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At simulation initialization, drones will appear at the map's center at their lowest altitude, point (50, 50, 1), since swarms usually start from the same point. They do not have any prior information about the map; they will explore and identify the characteristics of the map. There is communication between each drone, as they all contribute to a shared map. It is assumed that drones can only sense elements at or below their flight height, within a radius equal to that height. A representation of how they observe their environment is shown in Fig. 2.

When sensing obstacles, drones will identify an object with 100% accuracy, but they will only log what they can see. For example, if a drone is flying at z = 3 and there is an obstacle within its sensing radius at z > 3, the drone will only know the obstacle's position and that it is at least 3 units tall. Width is treated similarly.

When sensing targets, sensing accuracy depends on altitude: a high-altitude drone is less likely to identify a target than a low-altitude one. Probability of finding a target will be 100% at z =1 and 10% at z = 10, following:

$$P_z = \frac{1}{z} \tag{1}$$

As mentioned, drones are not able to sense through obstacles.

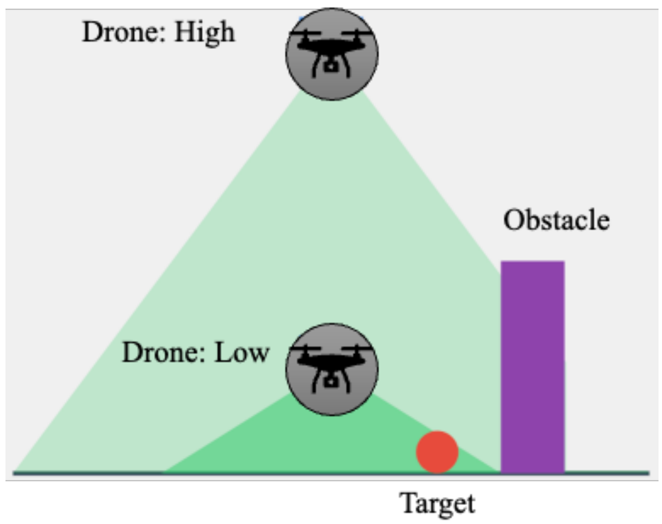

Figure 2: Representation of drone sensing abilities. The higher they fly the more area they can see, but their chances of identifying targets is lower. Drones cannot sense through objects.

B. Soft Actor-Critic Algorithm

SAC is a novel RL method that aims to optimize agent behavior in continuous action spaces [33]. It maximizes expected rewards while ensuring exploration through an entropy-based regularization term. It is composed of two main elements: actor and critic.

1. The actor or policy network

The actor is the decision-maker in each agent. It is trained to select actions that maximize the expected Q-value and entropy, which encourages exploration. Its objective function is as follows.

$$J_\pi(\phi) = E_{s\sim D, a\sim \pi_\phi}\left[\alpha \cdot log\pi_\phi(a \mid s) - Q_\theta(s,a)\right] \tag{2}$$

where $\alpha \cdot log\pi_\phi(a|s)$ is the entropy term and $Q_\theta(s,a)$ is the expected Q-value term. $E_{s\sim D,a\sim\pi\phi}$ is the expectation over states s in the replay buffer D, and over actions a in the current policy π_ϕ. The temperature parameter α fixes the balance between exploration and exploitation of well-known actions. The policy parameter updates happen continuously by using the gradient descent

$$\phi \leftarrow \phi - \eta_\pi \nabla_\phi J_\pi(\phi) \tag{3}$$

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where η_π is the learning rate for the policy network.

2. The critic or Q-function network.

The critic is the evaluator of the expected return on actions by the actor in a given state to enable the actor to improve its decision-making. It is trained by minimizing the Mean Squared Error between predicted Q-values, and target Q-values by the loss function

$$L(\theta) = E[(Q_\theta(s,a) - y(r,s'))^2] \tag{4}$$

where predicted Q-values are $Q_\theta(s,a)$. The target Q-values y(r,s') are computed in

$$y(r,s') = r + \gamma(Q_{\bar{\theta}}(s',a') - \alpha log\pi_\phi(a' \mid s')) \tag{5}$$

where r is the immediate reward after taking an action a in a state s, γ is the discount factor that balances the importance of immediate rewards versus future rewards, and $Q_{\bar{\theta}}(s',a')$ is the next state s' and next action a' estimate Q-value using the target network parameters. The discount factor r determines how the target network parameters track the critics' parameters. If it is one, the target network's parameters will be equal to the critics' parameters, and if it is 0, the target will not update. Lastly, the critic's parameters update in a similar manner to the actor's parameters

$$\theta \leftarrow \theta - \eta_Q \nabla_\theta J_Q(\theta) \tag{6}$$

where η_Q is the learning rate for the critic network.

3. Neural Network Architecture.

The architectures in the networks of both actor and critic follow the same structure composed of 3 dense fully connected layers with ReLU with the difference being in the size of their outputs as seen in Fig. 3. In the case of the actor, the outputs are the mean and the log standard deviation of the action distribution. They allow the computation of the objective function using

$$log_{\pi_\phi}(a \mid s) = Normal(\mu,\sigma).log\_prob(a) - log(1 - a^2 + \epsilon) \tag{7}$$

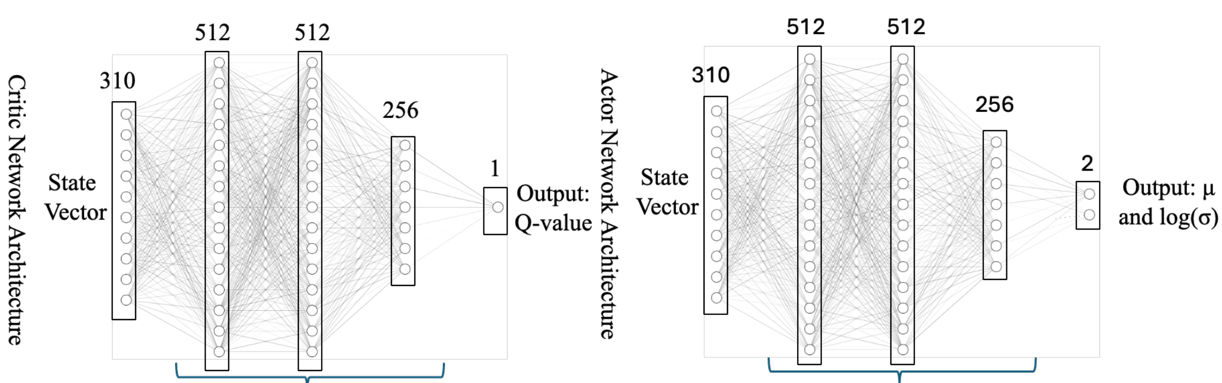

Figure 3: Neural Architecture of the Critic (left) and Actor (right) Networks.

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4. State Representation of the Environment

The state space represents the environment at any moment during the simulation, influencing agents' decision-making. In Table 1, there is a summary of all the components in the state vector, their dimensions, and their description.

- The drone position is presented in a 3D vector, and it is required for the drones to know their location within the operational space.
- Obstacle perception represents each drone's maximum sensing area as a grid of 10x10, as that is the area covered when it is flying at the maximum height of z = 10. The other 2 variables are a Boolean variable that indicates if a coordinate has an obstacle and a numeric variable that indicates each obstacle's maximum registered height. Maximum height and obstacle position are shared between drones via the shared map.
- Similarly, target perception is a 10x10 grid that indicates the presence or absence of a target in a coordinate.
- The relative position to the nearest drone is a 3D array that provides each drone's distance to the nearest drone. It helps in avoiding drone collisions and in allowing strategic formations, preventing clustering.
- A vector to the closest unexplored area aims to improve map exploration and avoid overanalyzing different sections.
- The remaining target count helps in making decisions based on the targets still to be found. If most of the area has been covered and there are still many targets unfound might suggest that those areas need to be inspected at a lower altitude.
- A step count will help drones make decisions considering time, as it is vital to operate efficiently in SAR missions.

Table 1: State Vector Composition

Component	Dim	Description
Drone Position	3	X, Y, Z coordinates of the drone
Obstacle Perception	200	10x10x2 grid representing obstacles and their heights
Target Perception	100	10x10 grid representing detected targets
Relative Position to Nearest Drone	3	Vector to the nearest drone
Vector to Closest Unexplored Area	2	Direction to the nearest unexplored area
Remaining Targets Count	1	Number of targets yet to be detected
Normalized Step Count	1	Current step / max steps (progress indicator)
Total	310	Total dimensions of the state vector

5. Action Space

The action space of the swarm of drones is relatively simple, as the only action that can be performed by each drone is to move in the 3D space. Velocity in this simulation is constant at one unit of distance per step.

6. Training rewards and penalties

To guide how the model learns, there is a need to specify different goals that will reward the system and some prohibited actions that, if performed, will incur penalties. These actions are presented in Table 2. The expected characteristics of the swarm of UAVs are that they will prioritize target detection, avoiding revisiting explored areas for increased efficiency; they will maintain distance with other drones to avoid unnecessary clustering; they will learn to avoid obstacles and respect boundaries, and will aim to finish the mission as soon as possible. To avoid great values entering the NN, which can lead to instability in training, all reward values are normalized after each step.

C. Implementation

The model is trained using Python 3.11 with PyTorch for the NN and optimization: NumPy for numerical operations, and Matplotlib for visualization. The model prioritizes GPU CUDA usage over CPU to accelerate training.

Training is carried out in 10 different maps for adaptability of different obstacle densities, as seen in Fig.1. The targets are assigned different locations every time a map is loaded, to avoid overfitting the designed maps.

Table 2: Reward and penalty system for training the model

Type	Value	Condition
Target Detection	150	Per target detected
Exploration	40	Per unexplored cell visited
Distance Reward	0.1 to 25	Based on average distance between drones
Obstacle Collision	-50	Per collision with obstacle
Boundary Collision	-50	Per collision with boundary
Clustering Penalty	-100	Per nearby drone (within 5 units)
Revisit Penalty	-40	For not moving
Time Penalty	-1	Per step
All Targets Found	10000	When all targets are detected
Early Termination	(max_steps - current_step) * 250	When all targets found before max steps

VI. Results

The training concludes when the model consistently finds over 80% of the targets in the training scenarios (i.e., after 50 simulations with over 80% of the targets found). The initially obtained model is then compared to two basic navigation algorithms to test its capacities: random walking and greedy exploration. The three models are tested on newly created maps, distinct from the training maps, over 50 runs. The tests were limited to the first two hours to evaluate the efficiency of each approach. The designed model outperforms the other navigation methods, as seen in Fig. 4. Not only are the average and median numbers of targets found by the SAC model across all runs higher, but it also finds targets faster than the other methods. Nevertheless, these results do not show a significant improvement over the simpler Random Walk algorithm, suggesting that the model may have stopped training too soon or overfitted in certain situations. Further training continues until the model consistently finds all targets within an hour, as shown in Fig. 5. This model shows improved performance on simpler algorithms, finding targets faster and more consistently, and is obstacle-aware. Other algorithms do not consider collisions or drone clustering. While it shows promising results, finding all targets within the hour in most cases, the results present high variance. In SAR, having this much performance variance does not justify implementing a model of this type. Known strategies, like creeping line, last-known-point spiral searches, do not require extensive training of an ML model and have proven to be very efficient in different real scenarios.

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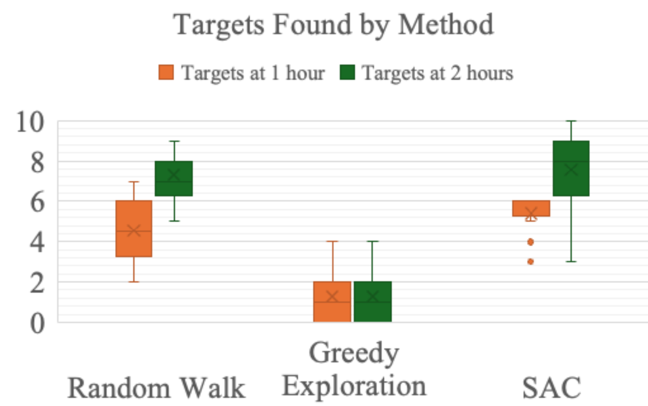

Figure 4: Targets Found by each method in 50 different scenarios in the first 2 hours.

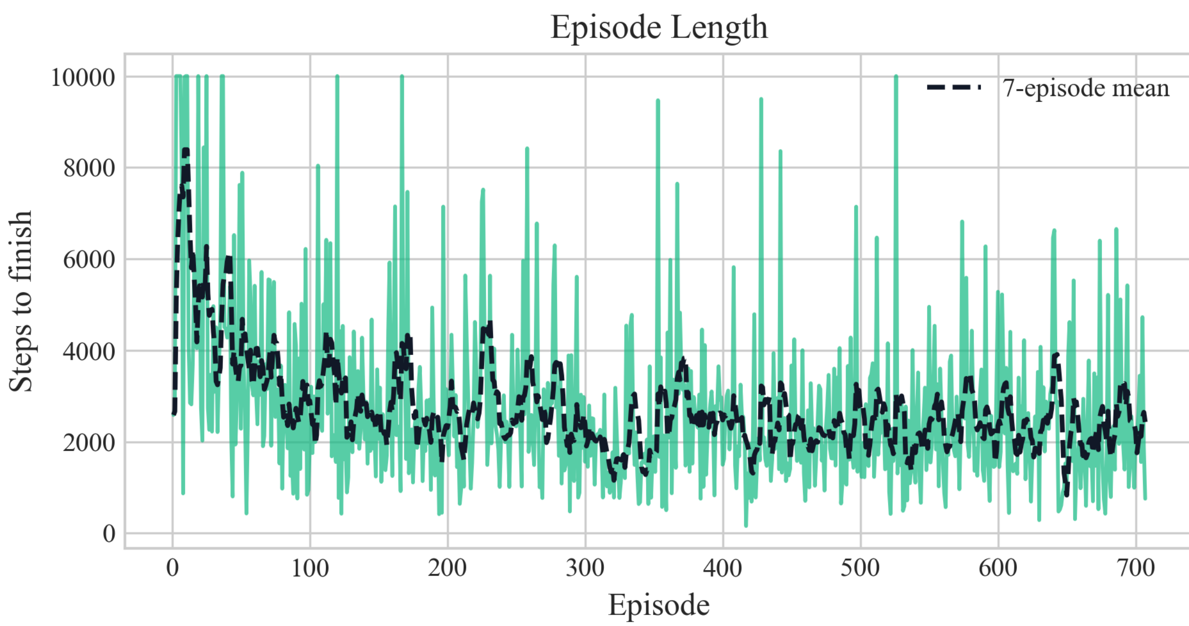

Figure 5: Episode steps to find all targets in training, until performance stops increasing.

VII. Conclusion

This study demonstrates the utility of applying Soft Actor-Critic (SAC) reinforcement learning to multi-drone search-and-rescue operations in complex 3D environments. With SAC, the drones were trained to efficiently navigate unfamiliar terrain, avoid obstacles, and coordinate with one another to maximize target detection. SAC is then shown to achieve better mission performance than random walking and greedy exploration navigation algorithms across different scenarios. However, despite these promising results, the model exhibits high performance variance, making its standalone deployment difficult to justify for critical operations compared to consistent, traditional methods. Consequently, future research should investigate Multi-Agent Proximal Policy Optimization (MAPPO) or hybrid architectures that integrate SAC with deterministic strategies (such as spiral or creeping line searches) to combine the adaptability of RL with the reliability of established patterns. Additional future work can focus on adding features to the model, such as fault detection in the swarm; combining it with other approaches to further improve performance; testing it in other scenarios or missions; or testing the model's scalability.

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